

The Data management module is mostly responsible for storing the patients' and staff members' information in the core server, and handle data retrievals from hospital staff.

The DataStorage class acts as the central repository, maintaining a map of patients and staff members, along with configurable alert thresholds. Each Patient holds a list of PatientRecord entries, where every record captures a measurement value, a record type (for example ECG blood pressure, blood oxygen saturation), and a timestamp (when the record was captured). Patients can add new records, retrieve records within a specified time range, and automatically remove records older than 180 days.

Access to patient data is controlled through the DataRetriever class, which enforces a minimum role level requirement before fulfilling any data request. When a staff member submits a query, DataRetriever verifies that the staff member exists in the active registry and holds a sufficient role level through validateAccess(). Every access attempt logged by the AuditLogger, which records DataRequest entries containing the patient ID, the staff member ID, and whether access was granted.

StaffMember objects carry an Id number and a role level, which determines their data access options. Together, these components form a role-based access control system.